

B-Type Natriuretic Peptide (BNP)

Test Details

METHODOLOGY

Immunochemiluminometric assay (ICMA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Support a diagnosis of congestive heart failure (CHF)

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Custom Additional Information

B-type natriuretic peptide (BNP), also referred to as brain natriuretic peptide, is a member of a family of structurally similar peptide hormones that includes atrial natriuretic peptide.¹ BNP is a 32-amino-acid peptide that contains a 17-amino-acid ring structure that is formed as the result of an internal disulfide bond. BNP is predominantly secreted by the ventricles of the heart in response to increased pressure. BNP acts as a vasodilator and has diuretic and natriuretic properties. BNP suppresses both sympathetic tone and the renin-angiotensin system. These physiologic effects serve to reduce intraventricular pressure and improve the symptoms of congestive heart failure (CHF).

BNP levels increase with age and can be transiently increased by vigorous exercise.^{1,2} BNP levels correlate with end-diastolic pressure and tend to be increased in patients with diminished left ventricular ejection fraction.^{1,3} BNP levels have been shown to be useful in the diagnosis of patients with symptoms that are consistent with CHF. Maisel and coworkers⁴ found that a BNP cutoff of 75 pg/mL could be used to predict systolic and/or diastolic abnormalities accurately in symptomatic patients referred to echocardiology for left ventricular function studies. They found that this cutoff identified patients with abnormal function from a population of 200 patients with a sensitivity of 95.9%, specificity of 86%, positive predictive value of 98%, and a negative predictive value of 89%. The overall accuracy of the diagnosis with a cutoff of 75 pg/mL was 93%. These results were similar to those found by Dao and coworkers.⁵ They found that BNP levels could be used to accurately identify patients with CHF from a population of patients presenting at the emergency room with shortness of breath (dyspnea). They found that a BNP value >80 pg/mL could be used to identify patients with CHF from this population with a sensitivity of 97%, specificity of 98%, positive predictive value of 92%, and a negative predictive value of 98%.⁶

Specimen Requirements

Specimen

Plasma, **frozen**

Volume

0.8 mL

Minimum Volume

0.5 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Lavender-top (EDTA) tube

Collection Instructions

Do **not** collect in glass. Collect in a plastic lavender-top (EDTA) tube. Centrifuge and transfer the plasma into a LabCorp PP transpak frozen purple tube with screw cap (LabCorp N° 49482). Freeze within 24 hours and maintain frozen until tested. To avoid delays in turnaround time when requesting multiple tests on frozen samples, **please submit separate frozen specimens for each test requested.**

Stability Requirements

Temperature	Period
Room temperature	Unstable (stability provided by manufacturer or literature reference)
Frozen	90 days (stability provided by manufacturer or literature reference)

Reference Range

0–100 pg/mL

Storage Instructions

Freeze.

Causes for Rejection

Non-EDTA plasma specimen; EDTA plasma collected in glass

Footnotes

1. Maisel A. B-type natriuretic peptide levels: A potential novel “white count” for congestive heart failure. *J Card Fail.* 2001; 7(2):183-193.
PubMed 11420771 (<http://www.ncbi.nlm.nih.gov/pubmed/11420771>)
2. McNairy M, Gardetto N, Clopton P, et al. Stability of B-type natriuretic peptide levels during exercise in patients with congestive heart failure: Implications for outpatient monitoring with B-type natriuretic peptide. *Am Heart J.* 2002; 143(3):406-411. **PubMed 11868044**
(<http://www.ncbi.nlm.nih.gov/pubmed/11868044>)
3. Valli N, Georges A, Corcuff JB, et al. Assessment of brain natriuretic peptide in patients with suspected heart failure: Comparison with radionuclide ventriculography data. *Clin Chim Acta.* 2001; 306(1-2):19-26.
PubMed 11282090 (<http://www.ncbi.nlm.nih.gov/pubmed/11282090>)
4. Maisel AS, Koon J, Krishnaswamy P, et al. Utility of B-natriuretic peptide as a rapid, point-of-care test for screening patients undergoing echocardiography to determine left ventricular dysfunction. *Am Heart J.* 2001; 141(3):367-374. **PubMed 11231433**
(<http://www.ncbi.nlm.nih.gov/pubmed/11231433>)
5. Dao Q, Krishnaswamy P, Kazanegra R, et al. Utility of B-type natriuretic peptide in the diagnosis of congestive heart failure in an urgent-care setting. *J Am Coll Cardiol.* 2001; 37(2):379-385. **PubMed 11216950**
(<http://www.ncbi.nlm.nih.gov/pubmed/11216950>)

6. Wu AH, Packer M, Smith A et al. Analytical and clinical evaluation of the Bayer ADVIA Centaur Automated B-type natriuretic peptide assay in patients with heart failure: A multisite study. *Clin Chem*. 2004 May; 50(5):867-873.